

We Could Soon Settle the Biggest Controversy in Maths

Key words and phrases from the New Scientist article · Level C

The words and phrases below appear in the article *We Could Soon Settle the Biggest Controversy in Maths* by Alex Wilkins, published in *New Scientist* (18 April 2026). The vocabulary spans three registers: mathematical and logical terminology, academic process language, and evaluative and descriptive expression. Study the table carefully before attempting the gap-fill exercise in Part 2.

PART 1 — VOCABULARY TABLE

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
conjecture	a mathematical statement believed to be true but not yet formally proved	<i>The ABC conjecture is simple to state but has resisted proof by the world's best mathematicians for decades.</i>
prime numbers	numbers greater than 1 that can only be divided by 1 and themselves	<i>The ABC conjecture concerns prime numbers involved in the solutions to the equation $a + b = c$.</i>
proof	a rigorous mathematical argument that demonstrates a statement must be true	<i>Mochizuki posted his 500-page proof online in 2012, claiming to have solved the ABC conjecture.</i>
theorem	a mathematical statement that has been formally proved to be true	<i>At 70 lines of code, you would struggle to prove a couple of undergraduate-level theorems.</i>
formalisation	the process of translating a written mathematical proof into a computer language that can verify it automatically	<i>There had been immense progress in formalisation — turning proofs into code that computers can check.</i>
verification	the process of confirming that something is correct or true, especially through systematic checking	<i>Mochizuki does not see verification as the central purpose of the Lean formalisation project.</i>
IUT (Inter-universal Teichmüller theory)	the new mathematical framework developed by Mochizuki to prove the ABC conjecture	<i>IUT featured entirely new techniques and concepts that most mathematicians found impossible to follow.</i>
bitterly contested	strongly and persistently disputed, with deep disagreement between opposing sides	<i>The correctness of Mochizuki's proof is one of the most bitterly contested questions in modern mathematics.</i>
on the verge of	very close to a particular point or moment; about to happen	<i>The controversy may be on the verge of being resolved, thanks to two new computer-based projects.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
bombshell	something unexpected and shocking that has a major impact on a situation	<i>Mochizuki's proof was a mathematical bombshell — astonishing the community but impossible for most to verify.</i>
impasse	a situation in which no further progress seems possible because of fundamental disagreement	<i>After Scholze and Stix identified a possible error, the debate reached a complete impasse.</i>
cadre	a small, tightly knit group of experienced people who share a belief or set of skills	<i>Mochizuki and a cadre of close colleagues at Kyoto University maintained that the proof was correct.</i>
naysayers	people who habitually express doubt, opposition, or pessimism about something	<i>Mochizuki offered an olive branch to his naysayers and suggested a potential path forward.</i>
olive branch	an offer or gesture intended to make peace or resolve a dispute	<i>Mochizuki offered an olive branch to critics by suggesting the proof could be formalised in Lean.</i>
helmed by	led or directed by a particular person	<i>One of the two formalisation projects is helmed by Mochizuki himself.</i>
intermediary	a person who acts as a go-between, carrying messages between parties who will not communicate directly	<i>Communication with Mochizuki happened only via an intermediary — he refused direct contact.</i>
enlisting	recruiting people to help with a task or cause	<i>Fumiharu started the LANA project by enlisting mathematicians familiar with Mochizuki's work.</i>
immune to	not affected by or protected against something harmful or undesirable	<i>Mochizuki says Lean formalisation will produce a record of IUT that is immune to misinterpretation.</i>
misinterpretations	incorrect understandings or explanations of something	<i>Mochizuki believes the controversy exists largely due to false misinterpretations of his work.</i>
far-reaching	having a wide range of effects or consequences, extending well beyond the immediate context	<i>Proving the ABC conjecture would have far-reaching implications for many other areas of mathematics.</i>
clarify	to make something easier to understand by explaining it more clearly	<i>Prominent mathematicians spent months trying to clarify Mochizuki's work without success.</i>
publicly available	open for anyone to access and inspect; not restricted or secret	<i>The 70 lines of Lean code produced so far are not yet publicly available to the wider community.</i>
settle	to resolve a dispute or question definitively, ending the uncertainty	<i>The primary goal of the LANA project is to settle the controversy once and for all.</i>
focal point	the central point of interest, attention, or activity in a situation	<i>Mochizuki says the verification aspect is not a central focal point of interest for his team.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
let alone	used to emphasise that something is even more impossible or unlikely than what was just stated	<i>At 70 lines, you struggle to prove basic theorems — let alone a 500-page mathematical proof.</i>

PART 2 — GAP-FILL EXERCISE

Complete the sentences using the correct word or phrase from the word bank below. You may need to change the form of the word. Most sentences require more than one answer. Pay close attention to the article's argument — students who have read carefully will have a clear advantage.

WORD BANK

conjecture	prime numbers	proof	theorem
formalisation	verification	IUT (Inter-universal Teichmüller theory)	bitterly contested
on the verge of	bombshell	impasse	cadre
naysayers	olive branch	helmed by	intermediary
enlisting	immune to	misinterpretations	far-reaching
clarify	publicly available	settle	focal point
let alone			

1	The ABC _____ is simple to state — it concerns _____ involved in solutions to $a+b=c$ — but solving it requires deep insights into how addition and multiplication interact, with _____ consequences for mathematics.
2	Mochizuki's _____ was a mathematical _____ in 2012 — astonishing the community with its 500-page length and its entirely new framework, known as IUT.
3	Prominent mathematicians attempted to _____ Mochizuki's work, but the debate over its correctness reached a complete _____ after Scholze and Stix identified a possible error.
4	Mochizuki and a _____ of close colleagues maintained the proof was correct, while his _____ in the wider community disagreed — and he offered them an _____ by proposing formalisation in Lean.
5	The new field of _____ allows written mathematical proofs to be translated into computer code, enabling automatic _____ of their correctness — a process that could finally _____ the long-running dispute.
6	The LANA project was _____ by Kato Fumiharu, who began _____ mathematicians familiar with IUT alongside experts in Lean who had already formalised other large mathematical projects.
7	Even after workshops and indirect communication via an _____, Fumiharu and his colleagues became stuck on the exact point that Scholze and Stix had first identified — they could make no further progress.

8	Mochizuki sees the value of the project less in _____ correctness and more in communication — producing a precise record of IUT that is _____ false _____.
9	The _____ of the whole formalisation effort, Mochizuki argues, is not to prove he was right but to produce a clear logical structure that others can finally follow and understand.
10	The project led by Mochizuki is not yet _____ — only 70 lines of Lean code have been produced so far, which is enough to prove a couple of basic _____, _____ a 500-page proof.
11	Kevin Buzzard acknowledged that the 70 lines of code are not much — but described them as some of the most significant progress in understanding the proof since it was first announced, because the whole controversy had been _____ in every sense.
12	Whether or not the proof is ultimately shown to be correct, the _____ implications of resolving the ABC conjecture — for number theory, algebra, and other areas of mathematics — would be profound and wide-ranging.

Write your answers here:

1. _____	2. _____	3. _____
4. _____	5. _____	6. _____
7. _____	8. _____	9. _____
10. _____	11. _____	12. _____